ELEC50011A - MATHEMATICS 2A

The residue of a complex function $F(z)$ at a pole $z = a$ of multiplicity m is given by the expression

$$\lim_{z\to a}\left\{\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m F(z)\right]\right\}.$$

1. Let a mapping from the $z-$plane to the $w-$ plane be given by

$$w(z) = \sqrt{2}e^{-i\pi/4}z - 2 + i).$$

A region R in the $z-$plane is bounded by the $x-$ and $y-$axes, and the lines $x = -1$, and $y = 2$. Determine the region R^* of the $w-$plane into which R is mapped under the transformation given by $w(z)$, explaining your reasoning. Sketch both regions R and R^*. [4]

Solution: A mapping of the form $w = kz$ maps lines to lines. Multiplication by $k = e-i\pi/4$ also represents a rotation around the origin by $-\pi/4$, so clockwise. Hence $e^{-i\pi/4}z$ is the original rectangular region, rotated clockwise by $\pi/4$, and multiplying by a real constant changes the lengths of the sides, so $\sqrt{2}e^{-i\pi/4}z$ is the rotated rectangle, with sides lengthened. Finally, adding $-2+i$ translates this rectangle by -2 horizontally, and $+1$ vertically. Picture at end.

2. A contour integral is given as

$$I \;=\; \oint_C \frac{e^{iz}}{z(z^2+1)}\,dz\,.$$

(a) Find the residues for the poles of $F(z) = \dfrac{e^{iz}}{z(z^2+1)}$ at $z = 0, i$, and use Cauchy's Residue Theorem to obtain I when the contour C consists of four parts: H_R, a semi-circular arc of radius R, in the upper half-plane, centred at the origin, the real intervals $[-R,-r]$ and $[r,R]$, and H_r: a semi-circular indentation of radius r, centred at the origin. You should *choose* H_r to be an indentation into the upper or the lower half-plane, and include a sketch of C. [3]

Solution: $F(z)$ has two simple poles at $z = 0, i$ with residues:

(i) at $z = 0$: Res $= \lim_{z\to 0} zF(z) = \lim_{z\to 0} \dfrac{e^{iz}}{z^2+1} = 1$;

(ii) at $z = i$: Res $= \lim_{z\to i}(z-i)F(z) = \lim_{z\to i} \dfrac{e^{iz}}{z(z+i)} = -\dfrac{1}{2e}$.

Two cases:
(*) if H_r is indented into the lower half-plane, the pole at $z = 0$ is included in C and Cauchy's Residue Theorem gives $I = 2\pi i$(Sum of residues at poles inside C) $= 2\pi i(1 - 1/2e)$
(**) if H_r is indented into the upper half-plane, the pole at $z = 0$ is excluded from C and Cauchy's Residue Theorem gives $I = 2\pi i$(Sum of residues at poles inside C) $= 2\pi i(-1/2e) = -\dfrac{i\pi}{e}$. Picture at end.

(b) Write I as a sum of four integrals, along the four parts of C, use Jordan's Lemma to show the integral along H_R vanishes as $R \to \infty$, and obtain the integral along *your choice* of H_r, as $r \to 0$. [4]

Solution: The integrals are

$$I = I_1 + I_2 + I_3 + I_4 \;=\; \int_{-R}^{-r} \frac{e^{ix}}{x(x^2+1)}\,dx + \int_{H_r} \frac{e^{iz}}{z(z^2+1)}\,dz + \int_r^R \frac{e^{ix}}{x(x^2+1)}\,dx + \int_{H_R} \frac{e^{iz}}{z(z^2+1)}\,dz$$

Given that all singularities are poles, and for the purposes of Jordan's Lemma $m = 1 > 0$ where the integrand is $\dfrac{e^{imz}}{z(z^2+1)}$, so it's sufficient that on H_R, $\left|\dfrac{1}{z(z^2+1)}\right| \approx \dfrac{1}{R^3} \to 0$, as $R \to \infty$, and we can apply Jordan's Lemma to assert that $I_4 \to 0$ as $R \to \infty$.

For I_2 on H_r we take $z = re^{i\theta} \Rightarrow dz = iz\,d\theta$ with two cases: (*) $\theta = \pi \ldots 2\pi$, as anti-clockwise, and (**) $\theta = \pi \ldots 0$, as clockwise. For case (*), the integral then becomes

$$I_2 = \int_\pi^{2\pi} \frac{e^{ire^{i\theta}} iz\,d\theta}{z(r^2e^{i2\theta}+1)}$$

and taking the limit $r \to 0$, we have $e^{ire^{i\theta}} \to 1$ and $r^2 e^{i2\theta} \to 0$, leaving only

$$\lim_{r\to 0} I_2 \;=\; \int_{\pi}^{2\pi} i\, d\theta \;=\; i\pi\,.$$

In case (**), only the limits of integration change and we have

$$\lim_{r\to 0} I_2 \;=\; \int_{\pi}^{0} i\, d\theta \;=\; -i\pi\,.$$

(c) Finally, use all the above results to obtain [4]

$$\int_{-\infty}^{+\infty} \frac{\sin x}{x(x^2+1)}\, dx\,.$$

Solution: As $r \to 0$ and $R \to \infty$ we join I_1 and I_3 on the real axis into one integral and have

$$I = \int_{-\infty}^{+\infty} \frac{e^{ix}}{x(x^2+1)}\, dx \;+\; \begin{cases} i\pi & case\ (*) \\ -i\pi & case\ (**) \end{cases} \;=\; \begin{cases} 2\pi i(1-2/e) & case\ (*) \\ -i\pi/e & case\ (**) \end{cases}$$

depending on case (*) or (**). Both give the same result:

$$\int_{-\infty}^{+\infty} \frac{e^{ix}}{x(x^2+1)}\, dx \;=\; \int_{-\infty}^{+\infty} \frac{\cos x}{x(x^2+1)}\, dx \;+\; i\int_{-\infty}^{+\infty} \frac{\sin x}{x(x^2+1)}\, dx \;=\; i\pi(1-1/e)$$

The integrand with cosine is odd, so the symmetric integral of an odd function is equal to zero, leaving

$$\int_{-\infty}^{+\infty} \frac{\sin x}{x(x^2+1)}\, dx \;=\; \pi\left(1-\frac{1}{e}\right)$$

3. Given the integral

$$I = \int_0^{2\pi} \frac{1}{\sin\theta - i}\, d\theta$$

show that

$$I = \oint_C \frac{1}{f(z)}\, dz$$

where C is the unit circle in the complex plane, and $f(z)$ is a quadratic polynomial. Hence, or otherwise, obtain I. [5]

Solution: On the unit circle $z = e^i\theta$ with $\theta = 0 \ldots 2\pi$. Hence we use $\frac{dz}{d\theta} = ie^{i\theta} \Rightarrow d\theta = \frac{dz}{iz}$ and $\sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right) = \frac{1}{2i}\left(z - \frac{1}{z}\right)$ and rewrite

$$I = \oint_C \frac{1}{\left[\frac{1}{2i}\left(z - \frac{1}{z}\right) - i\right] iz}\, dz\,.$$

and multiplying out we have

$$I = \oint_C \frac{1}{\frac{1}{2}z^2 + z - \frac{1}{2}}\, dz \;=\; \oint_C \frac{2}{z^2 + 2z - 1}\, dz \;.$$

The quadratic polynomial has roots $z = -1 \pm \sqrt{2}$, and $z_1 = -1 + \sqrt{2}$ is inside the unit circle, while $z_2 = -1 - \sqrt{2}$ is outside. So the integrand has a simple pole at z_1 with residue:

$$\lim_{z\to z_1} (z - z_1)F(z) \;=\; \lim_{z\to z_1} (z - z_1)\frac{2}{(z - z_1)(z - z_2)} = \lim_{z\to z_1} \frac{2}{z - z_2} \;=\; \frac{2}{z_1 - z_2} \;=\; \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}\,.$$

Finally, we use Cauchy's Residue Theorem and obtain that

$$I = 2\pi i(\text{Sum of Residues at poles inside } C) \;=\; (2\pi i)\frac{1}{\sqrt{2}} = i\pi\sqrt{2}.$$

4. A set of data corresponds to measurements of a quantity x at times t, as follows:

t	x
0	7
1	3
2	4
3	2

Find the line of best fit $x = at + b$ for the above data, using the method of least squares, with $a, b \in \mathbb{R}$ to be determined. [7]

Solution: The line of best fit has equation $x = at + b$, giving equations
$7 = a \cdot 0 + b$
$3 = a \cdot 1 + b$
$4 = a \cdot 2 + b$
$2 = a \cdot 3 + b$
In matrix form this is

$$A\underline{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \\ 4 \\ 2 \end{pmatrix} = \underline{\mathbf{b}}$$

The equation $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ is an overdetermined system with no solution, so we multiply by A^T for the least-squares method with normal equations

$$A^T A\hat{\underline{\mathbf{x}}} = A^T \underline{\mathbf{b}}$$

where $A^T A$ is invertible, given that A has independent columns, so a solution $\hat{\underline{\mathbf{x}}}$ exists. Hence,

$$A^T A\hat{\underline{\mathbf{x}}} = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{pmatrix} \hat{\underline{\mathbf{x}}} = \begin{pmatrix} 14 & 6 \\ 6 & 4 \end{pmatrix} \hat{\underline{\mathbf{x}}} = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 7 \\ 3 \\ 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 17 \\ 16 \end{pmatrix} = A^T \underline{\mathbf{b}}$$

As A has independent columns, $A^T A$ is invertible and the equation has solution

$$\hat{\underline{\mathbf{x}}} = (A^T A)^{-1} A^T \underline{\mathbf{b}} = \frac{1}{20} \begin{pmatrix} 4 & -6 \\ -6 & 14 \end{pmatrix} \begin{pmatrix} 17 \\ 16 \end{pmatrix} = \begin{pmatrix} -1.4 \\ 6.1 \end{pmatrix}$$

So the line of best fit is $x = -1.4t + 6.1$. Picture at end.

5. Prove that a set of n orthonormal vectors in $\mathbb{R}^n$ is a basis for $\mathbb{R}^n$. [4]

Solution: A set of n linearly independent vectors form a basis, so it's enough to show that orthonormal vectors are linearly indepdent. Let the set $\underline{\mathbf{q}}_1, \underline{\mathbf{q}}_2 \ldots \underline{\mathbf{q}}_n \in \mathbb{R}^n$ be an orthonormal set. Then $\underline{\mathbf{q}}_i^T \underline{\mathbf{q}}_j = 0$ for all $i \neq j$ and $\underline{\mathbf{q}}_i^T \underline{\mathbf{q}}_i = 1$ for all i. Let

$$x_1\underline{\mathbf{q}}_1 + x_2\underline{\mathbf{q}}_2 + \ldots + x_n\underline{\mathbf{q}}_n = \underline{\mathbf{0}} \qquad (*)$$

The vectors are linearly independent if the only solution to this equation is all $x_i = 0$. Multiply $(*)$ on the left by $\underline{\mathbf{q}}_1^T$:

$$\underline{\mathbf{q}}_1^T(x_1\underline{\mathbf{q}}_1 + x_2\underline{\mathbf{q}}_2 + \ldots + x_n\underline{\mathbf{q}}_n) = \underline{\mathbf{q}}_1^T\underline{\mathbf{0}} \Rightarrow x_1\underline{\mathbf{q}}_1^T\underline{\mathbf{q}}_1 + x_2\underline{\mathbf{q}}_1^T\underline{\mathbf{q}}_2 + x_3\underline{\mathbf{q}}_1^T\underline{\mathbf{q}}_3 + \ldots x_n\underline{\mathbf{q}}_1^T\underline{\mathbf{q}}_n = 0$$

Hence $x_1 \cdot 1 + x_2 \cdot 0 + x_3 \cdot 0 + \ldots x_n \cdot 0 = 0 \Rightarrow x_1 = 0$, given the property of the orthonormal set. Similarly, multplying $(*)$ on the left by $\underline{\mathbf{q}}_2^T$ we get $x_1 \cdot 0 + x_2 \cdot 1 + x_3 \cdot 0 + \ldots x_n \cdot 0 = 0 \Rightarrow x_2 = 0$, and so on. Hence all $x_i = 0$ is the only solution to $(*)$ and the n vectors are linearly independent, and hence a basis.

6. A matrix is given as

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} .$$

The columns of A form a basis for $\mathbb{R}^3$. Apply Gram-Schmidt orthogonalization to the columns of A to find an orthonormal basis for $\mathbb{R}^3$. [6]

Solution: Apply Gram-Schmidt. Let $\underline{\mathbf{a}}_i$ be column i of A. Then

$$\underline{\mathbf{x}}_1 \;=\; \underline{\mathbf{a}}_1 \;=\; \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \underline{\mathbf{x}}_2 = \underline{\mathbf{a}}_2 - \frac{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{a}}_2}{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{x}}_1}\underline{\mathbf{x}}_1 \;=\; \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

and we take $\underline{\mathbf{x}}_2 = (2,-1,1)^T$ to simplify the next calculation:

$$\underline{\mathbf{x}}_3 = \underline{\mathbf{a}}_3 - \frac{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{a}}_3}{\underline{\mathbf{x}}_1 \cdot \underline{\mathbf{x}}_1}\underline{\mathbf{x}}_1 - \frac{\underline{\mathbf{x}}_2 \cdot \underline{\mathbf{a}}_3}{\underline{\mathbf{x}}_2 \cdot \underline{\mathbf{x}}_2}\underline{\mathbf{x}}_2$$

$$= \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}{\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{6}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \;=\; \frac{1}{3}\begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix}$$

Now normalize to get the orthonormal basis $\underline{\mathbf{q}}_i = \dfrac{1}{|\underline{\mathbf{x}}_i|}\underline{\mathbf{x}}_i$

$$\underline{\mathbf{q}}_1 = \frac{1}{|\underline{\mathbf{x}}_1|}\underline{\mathbf{x}}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \underline{\mathbf{q}}_2 = \frac{1}{|\underline{\mathbf{x}}_2|}\underline{\mathbf{x}}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \underline{\mathbf{q}}_3 = \frac{1}{|\underline{\mathbf{x}}_3|}\underline{\mathbf{x}}_3 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} .$$

7. Given

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix},$$

its Singular Value Decomposition (SVD) is $A = U\Sigma V^T$. Show that the orthogonal diagonalization of the smaller of the two matrices AA^T or A^TA gives the singular values and one of the orthogonal matrices U, V. Then find the SVD, stating at each point the basis for a fundamental subspace of A. [10]

Solution: As A is 2×3 we require the smaller matrix AA^T which will be 2×2. Hence we substitute

$$AA^T = (U\Sigma V^T)(U\Sigma V^T)^T = U\Sigma V^T V\Sigma^T U^T$$

and as V is an orthogonal matrix, V^TV is the identity matrix, leaving $AA^T = U\Sigma\Sigma^T U^T$. The 2×3 matrix Σ has the singular values σ_i on the diagonal, so the 2×2 matrix $\Lambda = \Sigma\Sigma^T$ has diagonal elements σ_i^2. As U is orthogonal, we have $AA^T = U\Lambda U^T$, the orthogonal diagonalization of AA^T.

Now to find the SVD, we begin with

$$AA^T = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

and obtain its orthogonal diagonalization $U\Lambda U^T$. We find the eigenvalues solving $\det(A-\lambda I) = 0 \Rightarrow (2-\lambda)^2 - 1 = 0 \Rightarrow \lambda = 3, 1$. The normalized eigenvectors are found for $\lambda = 3$:

$$(A-3I)\underline{\mathbf{u}} = \underline{\mathbf{0}} \Rightarrow \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\underline{\mathbf{u}} = \underline{\mathbf{0}} \Rightarrow \underline{\mathbf{u}}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

and for $\lambda = 1$:

$$(A-I)\underline{\mathbf{u}} = \underline{\mathbf{0}} \Rightarrow \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\underline{\mathbf{u}} = \underline{\mathbf{0}} \Rightarrow \underline{\mathbf{u}}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

These are a basis for the column space of A, and give the columns of U so we have

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \Rightarrow AA^T = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = U\Lambda U^T,$$

the orthogonal diagonalization of AA^T. The singular values are the roots of the eigenvalues so

$$\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & \sqrt{1} & 0 \end{pmatrix}$$

To find $\underline{\mathbf{v}}_i$ we use $\underline{\mathbf{v}}_i = \frac{1}{\sigma_i}A^T\underline{\mathbf{u}}_i$ and get

$$\underline{\mathbf{v}}_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & -1 \\ -1 & 0 \\ 0 & 1 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} \text{ and } \underline{\mathbf{v}}_2 = \begin{pmatrix} 1 & -1 \\ -1 & 0 \\ 0 & 1 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

These two vectors are a basis for $C(A^T)$, the row space of A. Finally, the third column of V is a basis for $N(A)$, the nullspace of A and orthogonal complement of the row space, so we solve

$$A\underline{\mathbf{x}} = \underline{\mathbf{0}} \Rightarrow \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix}\underline{\mathbf{x}} = \underline{\mathbf{0}}.$$

A single row operation gives $z = y$ and $x = y$, so after normalization, we have

$$\underline{\mathbf{v}}_3 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow V = \begin{pmatrix} \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

The SVD is therefore

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} = U\Sigma V^T .$$

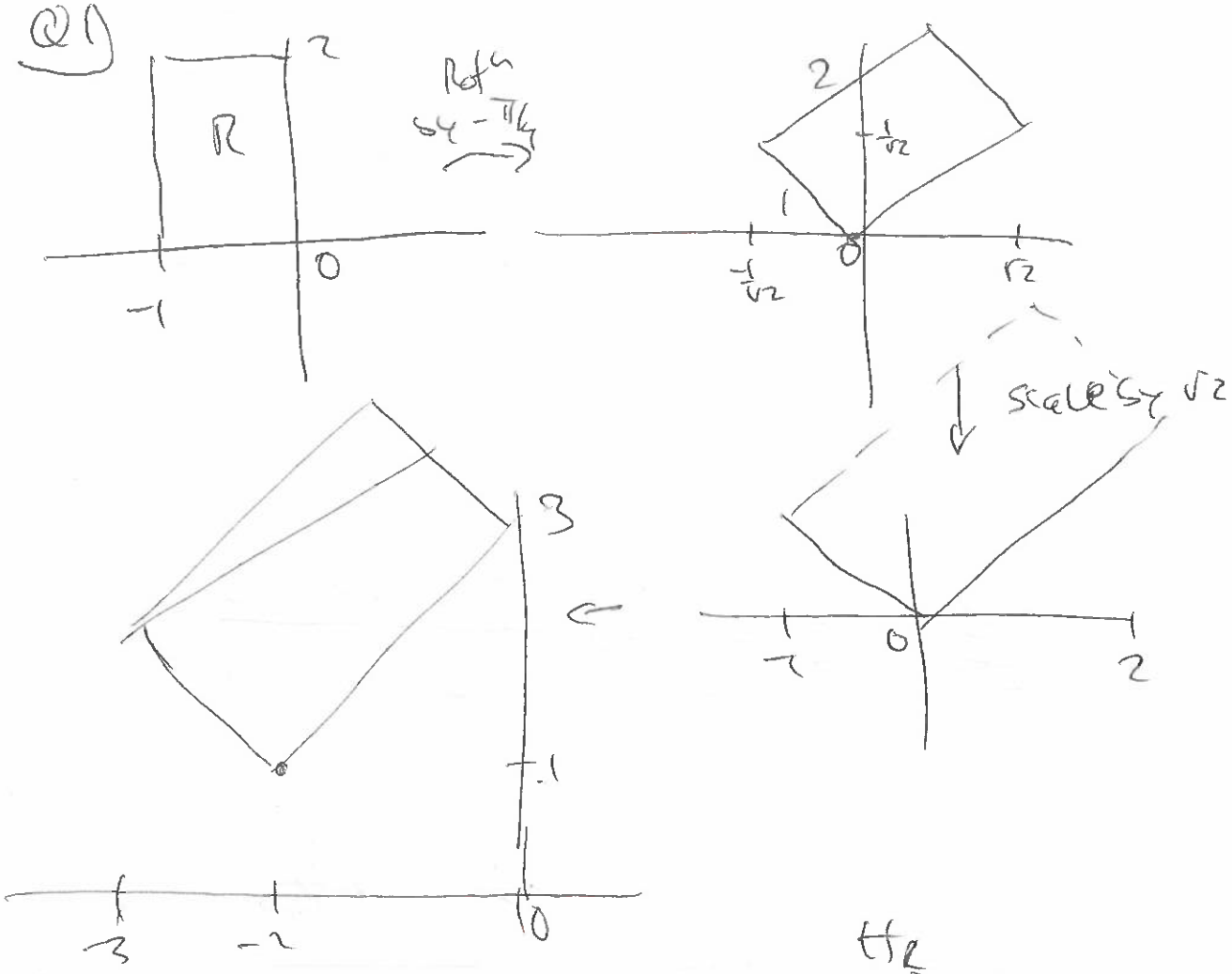
8. A digital image consisting of 2000×1000 pixels is compressed for transmission using the Singular Value Decomposition. The first r principal components are used, and the data transmitted is 15% of the original amount, to the nearest 1%. How many components were used? Explain your reasoning. [3]

Solution: The SVD is $A = U\Sigma V^T$, where A and Σ are 2000×1000 matrices, V^T is 2000×2000 and U is 1000×1000. Taking the first r components requires r columns each from U and V, and r singular values from Σ, a total of $r \times (2000 + 1000 + 1)$ transmitted data from the original 2000×1000 in A. As a percentage we estimate

$$\frac{r(3 \times 10^3 + 1)}{2 \times 10^6} = 1.5r \times 10^{-3} + 0.5r \times 10^{-6} \approx 1.5r \times 10^{-3}$$

to the nearest percent. Hence $1.5r \times 10^{-3} = 15\% \Rightarrow r = 100$.

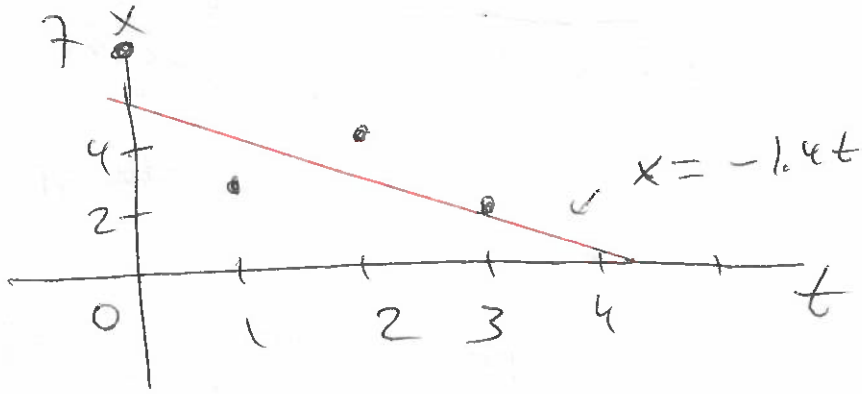
Q1)


R
2
-1
0
Rot^n
about -π/4
2
1/√2
1
-1/√2
0
√2
scale by √2
3
-1
-3
-2
0
2
-2
0

Q2) a


½R
-R
-r
r
R
½r
or
½R
-r
0
r
-R
R
½r

Q4)


x
7
4
2
0
1
2
3
4
t
x = -1.4t + 6.1